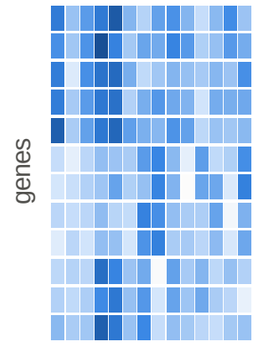
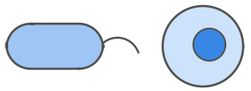
Adding a softmax and a multi-step walk to CLR's normalization recovers large *E. coli* regulons better (16% over CLR), but loses resolution on tight protein complexes

CLR (Faith et al. 2007)

1 Expression compendium



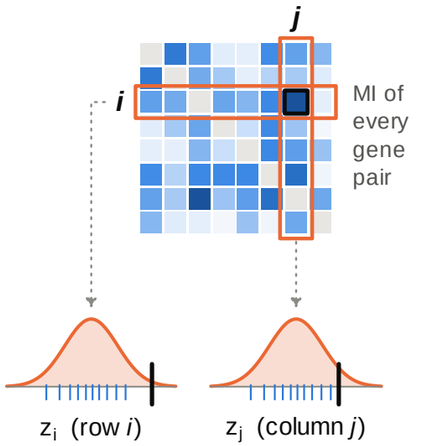


M3D, EcoMAC microarrays

K562 Perturb-seq

2 Mutual information, z-scored against each gene's own background

existing method



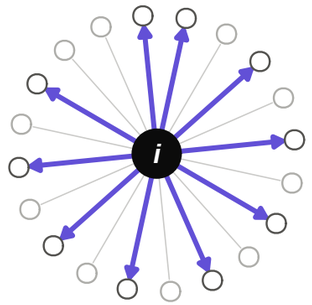
Stouffer combination; z-scores clipped at 0

Euclidean $\sqrt{z_i^2 + z_j^2}$ ranks the same

CLR score for the pair i–j

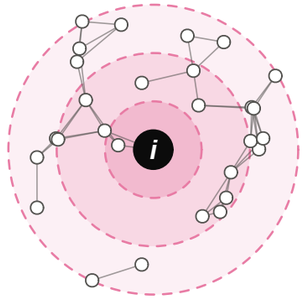
This work: steps taken from attention

3 Softmax with per-gene temperature



~10 effective partners per gene, set in advance

4 Multi-step walk, number of steps chosen without labels



steps t (log scale)



t = 9: chosen on held-out labs

Published CLR stops here: a score for each gene pair

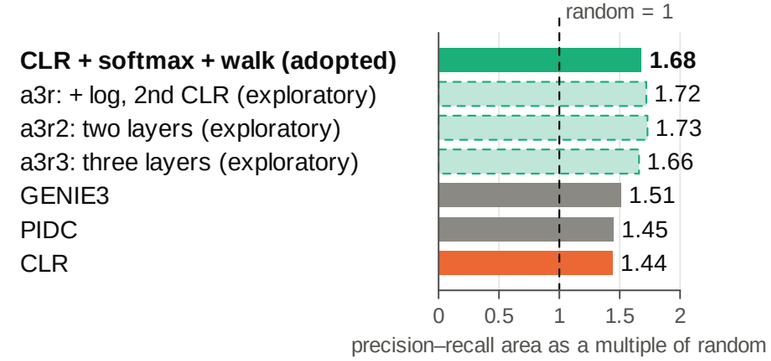
No learned parameters

Transformer analogy: MI ≈ query·key score; CLR z-score ≈ scaled logit; softmax ≈ softmax; walk ≈ repeated attention with residual

5 What it recovers

E. coli regulons (experimentally supported, RegulonDB)

regulon = genes controlled by one transcription factor (TF)



per regulon: tie with CLR and GENIE3; ahead in 1,000/1,000 resamplings vs CLR

Protein complexes (K562 CORUM; mostly ribosome): this work loses 1.8×

complex = proteins that bind as one machine

